AUTOMATED DUAL AXIS SOLAR TRACKING SYSTEM FOR A SMALL-SCALE PHOTOVOLTAIC (PV) PANEL

EEE381 Individual Investigative Project

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Abstract

This dissertation designs and validates an automated dual-axis solar tracking system for a small-scale photovoltaic (PV) panel. This system aims to enhance the performance and power generation of the PV panel by maintaining optimal solar alignment. Using an Arduino Uno, servo motors, and NOAA's solar position algorithm, the system achieved a 26% power increase over static panels, with angular accuracies of 3.37° (azimuth) and 0.92° (elevation) in calculations and ±0.57° in physical actuation. Key innovations include a 3D-printed servo bracket and proto-shield circuitry, demonstrating a cost-effective, scalable solution for improving solar energy capture. The results validate the potential of algorithm-based tracking, with future work suggested for hybrid sensor-algorithm designs and real-world testing.

Individual Contribution

During the course of this dissertation, the student has worked to conduct all of the system's development on their own. The system has a series of hardware and software which has been carefully selected by the student and has been assembled and programmed by the student with the use of libraries to more efficiently complete the system's program. The student has done the research of the project including its feasibility and its resulting effectiveness through the information sources from research papers and articles. This research has since been used in the development of the system while acting as comparative pieces to the testing results obtained from the system.

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To my parents and sisters for their endless support, motivation and understanding throughout the challenges faced throughout the conducting of this work. A special thanks to my mother who has worked tirelessly to support my education and being my biggest supporter. Without her, I would not be where I am and who I am today.

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Finally, I acknowledge the foundational contributions of researchers cited in this dissertation, whose pioneering work in solar tracking systems informed my approach. This acknowledgement extends to the open-source community for their contributions towards making complex components more accessible to all users.

Chapter 1

1. Introduction

1.1 Background and Motivation

Solar power is a renewable energy source that has come into high demand over the last 2 decades. It is a front runner of renewable energy, having become one of the fastest growing energy sources with an average annual growth rate of 24% over the last 18 years [1]. Solar power offers advantages such as versatility and an inexhaustible nature [2]. However, a notable limitation of this energy source is its efficiency which is lower than 25% at industry standard [3]. Throughout its years of development, the industry has taken considerable measures to improve the efficiency of solar cells whether that be by altering the semiconductor material, utilizing different layers of photovoltaic materials or simply optimizing solar alignment. These methods effectively address this issue and motivate this project, which seeks to design a system enabling a photovoltaic (PV) panel to maintain optimal solar alignment regardless of the time of day and position of the Sun. The PV panel is to be redirected to its optimal angle of sunlight exposure with a dual-axis solar tracking system which maximises the PV panels power output. This system is able to cover the Earth's rotation of its own axis and its full orbit around the Sun. This is a scale model project where the real-life applications would help advance the solar energy industry and its reliability as an alternative energy source to fossil fuels.

1.2 Problem Definition

This project aims to build an automated dual-axis solar tracking system for a small-scale photovoltaic (PV) panel.

1.3 Theory

The system applies a large variety of different theories to be able to achieve its functionality. This consists of different mathematical models and theories which determine the location of the Sun relative to the position of the system depending on the time and date of the calculation. Upon the determination of the position of the Sun, applications of system

programming allow for automated adjustment of the PV panel towards a tracked solar position. The applications of such principles are aiding factors in the concept of photovoltaics (PV) as light intensity is a dependent variable of a PV panel's power generation. [6]

Mathematical theory forms an essential foundation for the project's functionality. The main calculated variables to be found are the azimuth and elevation angles which are depicted in Figure 1. These variables are the respective relative horizontal and vertical angles of set longitudinal and latitudinal coordinates to the solar position. To obtain these angles, many sub calculations are required such as fractional years, equation of time, and declination angle [4]. These will all be highlighted as the theory is continuously discussed. These calculations are shown differently across different sources, but considering the accuracy of the calculations will be assessed relative to the National Oceanic and Atmospheric Administration (NOAA) solar position calculator, the source of mathematical models have been fully based on NOAA sources [4].

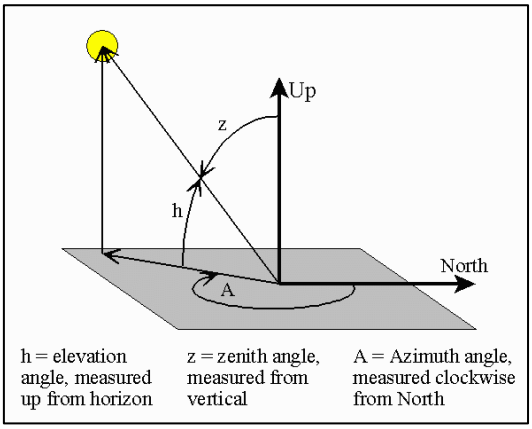

Figure 1 Depiction of Azimuth and Elevation Angles relative to the Sun [5]

Angular variables are critical in mathematical computation, as many calculations involve degrees or radians. This is an important aspect that were kept constant to ensure calculations are done accurately.

The system is to be fully automated which requires efficient and effective system programming. In this project, an Arduino Uno has been selected as the microcontroller that will automate the system’s functions. The Arduino Uno is a microcontroller board with its

own set of code libraries which allows it to compute data and signals to accurately output signals that help control components and devices. The Arduino Uno is user friendly and able to process the low data volume of this project's function. The code libraries available to the Arduino Uno help reduce the difficulty of programming individual components such as inbuilt serial read and write functions, and even component specific libraries such as for the servo motors and real-time clock. The mathematical model for solar position is translated into Arduino code and executed by the Arduino Uno where all the data will be processed. Planning the development of the programme is important towards its effectiveness and this is best done with a visualisation of the processing involved which will be expanded upon in the methodology.

Conceptualizing photovoltaics and their operation helps understand the need for such a project. Photovoltaics are semiconductor materials, typically silicon, which convert light energy into electricity. This energy produced by a PV cell is dependent on the size, conversion efficiency, and light intensity [6]. The project takes upon one of these main concepts contributing to the energy produced by the PV, that by invoking a direct path of sunlight towards the PV panel will maximise light intensity and solar energy absorption.

Efficiency is an outstanding concern of PVs as industry standards are still standing at an average lower than 25% [3]. Photovoltaics are being developed to further increase their efficiencies with several approaches being taken to do this. As the development of photovoltaics with multiple layers of different semiconductor material with ranging band gaps which allow a wider range of photon wavelengths to be absorbable energy which is harnessed into electrical energy [16]. However, maximising efficiency this way is expensive and is limited by the Shockley-Queisser limit which sets a theoretical maximum PV efficiency of 33.7% [7]. This development only progresses as a result of technological advancement and prompts the need for other paths of progression in which the solar tracking system becomes a potential solution.

An automated solar tracking system aims to improve the efficiency by aiming to maximise the light intensity which in turn maximises the usage of available solar energy [8] as shown by Figure 2.

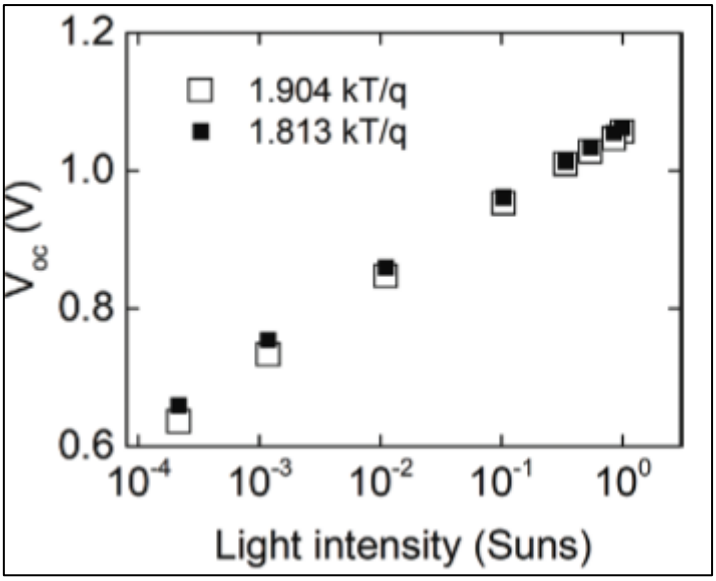

Figure 2 Photovoltaic Panel Open Circuit Voltage dependent on Light Intensity [8]

As depicted in Figure 2, the trend of Open Circuit voltage is directly proportional to light intensity which brings the need to maximise solar absorption. The feasibility and usability of this will be explored in the literature review as previous similar studies will be used to understand the work that has been done and the issues that were faced throughout that research.

1.4 Aims and Objectives

The project aims to develop a small-scale automated dual-axis solar tracking system which can direct a PV panel towards the Sun's direction in an effort to maximise solar energy absorption. This system will demonstrate the potential for increased efficiency of PV panels through optimised solar energy absorption. This aim will be achieved through the incorporation of the concepts of mathematical modelling, programming, circuit design, and hardware design.

The objectives outline the project specifications and the criteria for measuring its success. These objectives can be split into different categories that being design, system development, validation, and performance analysis.

The design objective is to develop an electromechanical concept which achieves the dual-axis solar tracking system's full range coverage of solar azimuth and elevation angles. This will be

the foundation of the project where the system's calculation of solar azimuth and elevation angles must accurately direct the PV panel towards the Sun where the dual-axis actuator mechanism must reflect this accurately.

Hardware development will aim to properly integrate all electronics, motors and mechanical components into a prototype with an automated control system. This will demonstrate the solar tracking system concept and serve as the foundation for future improvements.

Validation will test the tracking system within controlled environments to measure the accuracy and effectiveness in maintaining optimal solar alignment. There are 2 variables to validate solar tracking accuracy, that being the solar position calculations and actuator movement on both axes. Solar position calculations will be tested against the National Oceanic and Atmospheric Administration (NOAA) solar calculator to within ±5° for the azimuth angle and ±2° for the elevation angle. The accuracy of actuator movement will be measured against the solar calculations where the measurement of angle will be done with a circular protractor as shown in Figure 3.

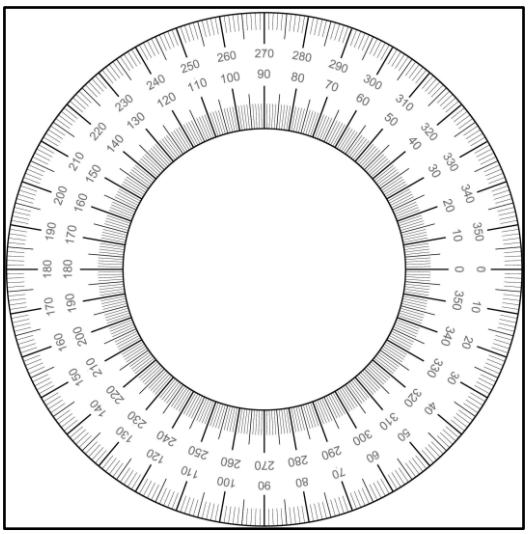

Figure 3 Picture of Circular Protractor [9]

Performance analysis will focus on how the system is able to enhance the performance of the PV panel and its efficiency. This will be achieved by simulating the PV panel with and without the solar tracking system. Experimentation will be done with a comparative simulation of a day of usage between the solar tracking system and the static solar panel which will be expanded further in the methodology. This compares the power generation and

exhibits the impact of the tracking system towards maximizing efficiency. From the literature review, prior research has shown 108.36% [10] higher power output in solar tracking systems compared to static solar panels. With this knowledge, targeting an increase of 20% power generation is a significant and reasonable difference in performance which this system will aim to prove.

These objectives and goals are listed in the project specification. Project Specification:

- Full Dual axis angle coverage of Azimuth and Elevation
- Operational integrated system of all components
- Comparative accuracy of azimuth angle calculation to NOAA solar calculator within ±5°
- Comparative accuracy of elevation angle calculation to NOAA solar calculator within ±2°
- Comparative accuracy of solar tracking system azimuth angle to calculated azimuth angle within ±1°
- Comparative accuracy of solar tracking system elevation angle to calculated elevation angle within ±1°
- Solar tracking PV panel to improve static PV panel power output by 20%

Chapter 2

2. Literature Review

Photovoltaics (PV) and their efficiencies are majorly dependent on their ability to capture sunlight. In real applications, the sunlight captured varies throughout the day due to the Sun's movement. Typical PV panels are structured statically in an array of panels which make them cost effective and uniform. However, this form of structure limits the ability of the panels to maximise their exposure to sunlight and reduces the light intensity they receive due to a lack of optimal solar alignment. To address this issue, solar tracking systems have been developed to enable PV panels to automatically adjust their orientation, maximizing sunlight exposure.

Solar trackers can be designed as single-axis or dual-axis systems, both achieving similar objectives with varying levels of success. A single-axis solar tracking system will only orient the PV panel either horizontally or vertically. The dual-axis system, as the name pertains, is able to operate at higher precision by tracking both the azimuth and elevation angles as it enables a full range of orientations [11]. The single-axis system is a simplistic and effective method which can adequately improve the sunlight exposure of the PV panel, but the complex and superiorly accurate dual-axis system truly maximises the exposure and efficiency of the PV panel. As the project focuses on the dual-axis system, the literature review will look towards other dual-axis solar tracking systems and how they have advanced the significance of PV panels and solar energy in the energy industry.

There is debate whether sensorless solar tracking systems are more efficient or utilizing sensors would be more efficient. This topic has been found to be in favour of sensorless systems, particularly in poor weather conditions as it was observed that the sensorless system was 4.2% more efficient than a solar tracking system with a light dependent resistor sensor [12]. This result can be drawn to several elements such as the increased number of external factors that would affect the tracking as other light sources may interfere in an outdoor environment.

Furthermore, the overall system may hint towards this result as more components being utilised would invoke a higher demand of power and therefore a lower overall efficiency. Although this study favours an algorithm-based approach, it is important to consider how weather conditions across different environments and project scale may favour a sensor-based system instead.

The research into this topic has become extensive over the last decade as many research papers have been published discussing the concept. Throughout the research completed, the aim has been to maximise the efficiency of PVs with solar tracking systems which have been shown to increase PV efficiency by 108.36% from static PV systems [10].

However, it is also important to consider factors that cause the implementation of such a system to be a liability and detrimental to its overall applicability. A significant factor is the powering of the automated control and actuator system as this affects the overall efficiency of the system as the movement of PV panels, especially as it enters larger industrial sizes. This can be highly power consuming and efficient movement is key towards the improving the true efficiency of these solar tracking systems.

Considering the factors of minimizing the system's power consumption from PV panel movement, several methods can be taken to achieve this such as limiting range of motion, PV panel weight optimization, and control system efficiency. Research into this topic of optimised movement discloses the influence of movement across different times of day on the power consumption of the system. This is a result of the angles at which the panel is being moved where angles close to 0° showed higher power consumption, contrary to angles close to 90° which tended to lower levels of power consumption [14]. For this small-scale project, the most significant factors to be considered are the use of high efficiency actuators and optimizing the solar tracking algorithm to balance movement frequency and power generation.

Taking a critical approach, it is evident of the lack of research for a hybrid of sensors and algorithm solar tracking system which could possibly take the advantages of both systems and mitigate each systems' weaknesses. However, it would be a more complex system as more data types are obtained and integrated together. There is also the consideration of more power being applied in powering the additional components that would also deteriorate the system's true efficiency. This is an outstanding factor in forgoing this hybrid system for this project as time constraints limit the overall success of the project while introducing a higher uncertainty of true efficiency as the subsequently increased power usage of the system will not be accounted for.

Across all sources used, it can be concluded that the automated dual-axis solar tracking system based on solar calculation algorithm is the ideal system to develop towards maximising tracking accuracy and PV solar energy absorption. This is because of its enhanced accuracy and reliability towards improving direct sunlight exposure and power generation. This method aims to ensure a high level of power generation to balance the power consumption of the tracking system in an effort to achieve a higher true efficiency from the PV panel.

Chapter 3

3. Methodology

3.1 Solar Tracking System Design

The Solar Tracking System is an electromechanical system which integrates a series of hardware and software components. The system's large range of components are the essential hardware tools of the project and are listed as shown in Table 1.

Component Type & Name	Component Type & Function
Servo Drives (x2) - Mini 180 Resin Gear Servo Drive	Used to direct the solar panel towards the tracked position of the Sun on the horizontal and vertical axis
LCD Display - Midas 0.96in Yellow Passive matrix OLED Display 128 x 64pixels COB I2C Interface	Displays the values of variables such as the Time, Date, Coordinates of Solar Panel, Power Generation, Azimuth & Elevation
Real Time Clock - Digilent 410-218, Pmod RTCC	Device keeps track of the time and date even when the system is shut down as it has its own external power source
Microcontroller - Arduino Uno Rev 4 WiFi	Microcontroller that processes all the Solar Tracking System's functions
Proto Shield - Arduino, Proto Shield Rev3 (Uno Size)	Proto Shield will be the binding piece for all the electrical and electronic components
Solar Panel - Seeed Studio 0.5W Kit solar panel	Used to test the effectiveness of the Solar Tracking System by giving numerical power generation values

Table 1 List of Components and their Individual Functions

Each component serves their own individual functions and varies in terms of control and application. These components have their own datasheets on the table in Appendix 14. Defining a system block diagram of the system's functionality helps act as a guide for the system's components. This block diagram is outlined in Figure 4.

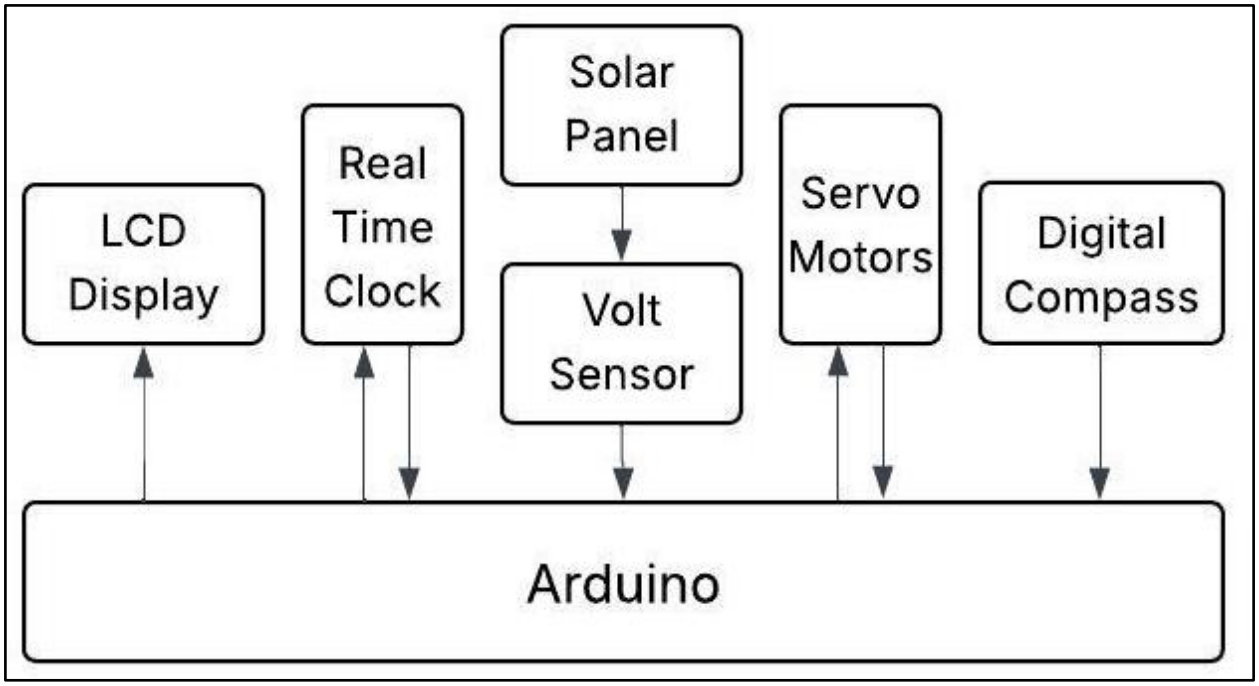

Figure 4 System Block Diagram

At a glance, Figure 4 outlines all the components involved in the solar tracking system and their communication with the microcontroller, the Arduino Uno. The Arduino Uno is in control of the solar tracking system functions such as processing calculations, adjusting the servos and communicating data to the LCD display to show measurements and calculations. Figure 4 uses arrows to describe the direction in which data is being communicated between the components and the Arduino Uno where in some components only communicate to the microcontroller, some which only receive data from the Arduino, and some do both.

With this system block diagram, the components and their data communication streams are defined which will help when designing the flow diagram of code for the microcontroller's programme.

3.2 Solar Position Calculation Code Development

The code of the solar tracking system is the fundamentals of its basic operation such as the calculation of solar position and control of the servo motors. A flow chart of the programme is an important part of ensuring basic functions of the system are correctly executed before looking to expand and improve the total function of the system. The flow chart of the programme is shown in Figure 5.

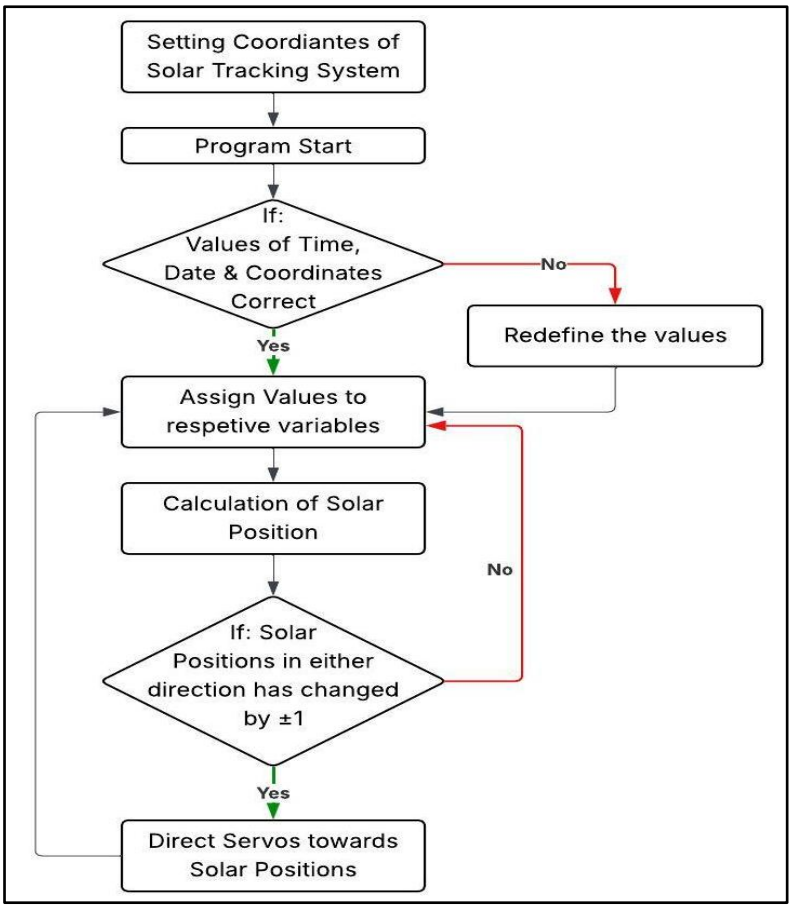

Figure 5 Flow Chart of Solar Tracking System Code

Figure 5 shows a series of conditional statements and functions which are denoted by diamonds and rectangles respectively. The code describes the Date and Time input from the Real Time Clock and is used to calculate the solar position with NOAA calculations. The servos are then adjusted if a ±1° change from its current position is observed.

This describes the overall program of the Solar Tracking System which is fully coded in the Arduino IDE and shown in Appendix 1. This code spanning over 700 lines performs the full function of the Solar Tracking System including the set up and operation of the Real Time Clock, LCD Display, and Servo Motors. The solar position calculation from Section 1.3 was implemented in Arduino C++ using NOAA's equations for azimuth/elevation. Achieving correctness within the code was done through the testing with the hardware mentioned in Table 1 which needs to be assembled.

3.3 Hardware Assembly

The hardware involved in this project spans beyond those shown in Table 1 as there are other parts that allow the system to be physically assembled. The main hardware parts outside of the components are the servo motor pan and tilt bracket and the Arduino Proto Shield. These pieces are essential in integrating all the components.

The servo motor pan and tilt bracket house the servo motors which allow for rotation across the x and y axis to translate to the azimuth and elevation angles respectively. This is best shown in Figure 6 where a depiction of the range of motion of the servo bracket is shown.

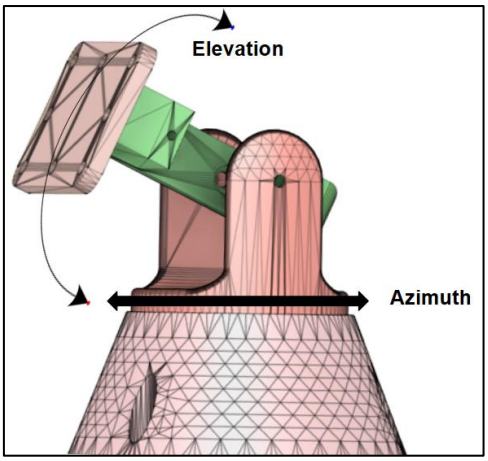

Figure 6 Diagram of Pan and Tilt Motions of Solar Tracking System

The initial servo bracket used can be found in Appendix 2 which was bought from RS Electronics which acted as an initial prototype for the system to verify the system's physical functionality. This was then changed to a 3D printed model which is shown in Appendix 3. This was a revised version where the design was made to be sturdier and hold the servos in place as the prior version struggled to hold the system in place when in movement. The design in Appendix 3 presents a larger base which holds the system in a more stable position when the system is active.

The Arduino Proto Shield acts as a binder for all the components involved in the Solar Tracking System as a breadboard does. In the initial prototype, the breadboard was used as shown in Appendix 4 to test the circuitry to ensure that all of the components are operational individually and in a fully integrated system.

As the breadboard was able to prove the functionality of the circuitry, this commenced the next stage of the circuit assembly with the Proto Shield. The Proto Shield requires soldering of components and wires which would make a more permanent connection of circuitry in comparison to the connections of a breadboard. Similarly to a Printed Circuit Board (PCB), it introduces a stable connection and structure to the system while cutting on its form factor as well. Appendix 5 shows the fully assembled Proto Shield with the subsequent components of the system.

Finally, the solar tracking system's hardware is assembled where system integration can commence as the product starts taking its final form.

3.4 System Integration

System integration is the merging of operations between the hardware and software elements of the project. In the case of the Solar Tracking System, the Arduino code is used to program the hardware elements of the system such as the servo motors and LCD display.

Throughout the development of the Arduino code, the code had been tested in applications with their corresponding components. This allowed the components to function with their respective programs before the full compilation of code was done. As software integration worked towards this large compilation of code, errors would take place in the case of identically named variables overlapping which required troubleshooting for the code to be functional.

The completed software integration phase allowed all the components to work simultaneously in theory, but the connections of the components were another challenge to tackle. With some components using the same pins such as SDA and SCL, some were required to share these pins which generally were not an issue as it was a multi-device accessible connection.

As the connections were made and the system was run with the full compilation of code, the solar tracking system was successfully integrated and all components were able to perform their functions as they did before full system integration. This marked the conclusion of the development of the system as it was ready for the beginning of the testing phase where the finalized system can be found in Appendix 6.

Chapter 4

4. Solar Tracking System Testing and Results

4.1 Testing Methodology

The Solar Tracking System was tested for 3 different aspects which were solar position calculation accuracy, physical positional accuracy, and solar tracking performance effectiveness. These tests were all conducted to ensure the solar tracking system would positively affect the PV panel's performance as poor execution would deter it. To quantify the results of these tests, a target specification has been set for each test outlined as outlined in the project specification.

Firstly, the solar position calculations were tested to ensure that calculations were accurate to that of the standard NOAA solar position calculator which is found in Appendix 7. This was an important aspect to test as this ensured that the directed solar panel was in the correct position as poor calculations could direct the solar panel away from the Sun and hinder its performance. This testing was done with a simulation of timestamps in the day (10:00, 12:00, 14:00, 16:00) once a month throughout the year 2023. These calculations would then be interpreted by both the standard NOAA calculator and the compiled mathematical algorithm which would be compared to observe the difference in these calculations. The target specification of this test was less than a 5° and 2° difference between the standard NOAA solar calculator and calculations of azimuth and elevation respectively. This ensures that the solar position is accurate and positively influences the performance of the PV panel.

Subsequently, the physical position of the solar tracking system is tested through the azimuth angle range of 270° and elevation angle range of 90°. This range was tested with a protractor as shown in Appendix 8 and 9 where the azimuth and elevation were tested in steps of 5° and 3° respectively. This test was done to validate the accuracy of the actuators used to direct the PV panel toward the Sun with a similar intention to the calculation test to ensure a positive effect on the PV panel’s performance. For this test, a specification of 1° would be a feasible and meaningful target to qualify a high level of accuracy for both azimuth and elevation.

Finally, solar tracking performance effectiveness testing is done to simulate the movement of

the Sun across a static PV panel and compare it to a solar tracking PV panel. This idea of testing is used as it is difficult to test this small-scale system outdoors as the small panel requires minimal light intensity outdoors to achieve its maximum power output. Furthermore, external factors made testing outdoors unreliable. Nonetheless, this test will determine the maximum generation angle range from the direct line of sight to the light source. The specification for this test aims for an increase of power output of 20% which is significantly smaller than the 108.36% increase from prior research [13]. However this specification is justified by the small scale level of this project and limitations to experimentation. Power output values above 5% of the maximum power output will be used to classify the maximum generation angle range which would classify angles with power output above 0.475W as maximum power outputs. Execution of this test was difficult to do with a consistent and accurate way of moving the light source across a static PV panel as shown in Figure 7.

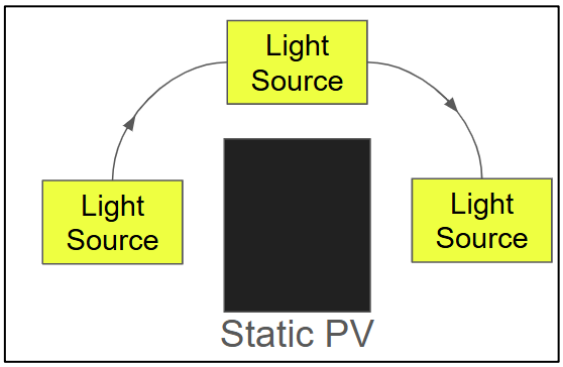

Figure 7 Diagram of Light Source across Static PV

To simulate the movement of the Sun more accurately across a static PV panel, the light source was left in a static position 10 cm away from the PV panel which moves through a 180° travel path as described in Figure 8.

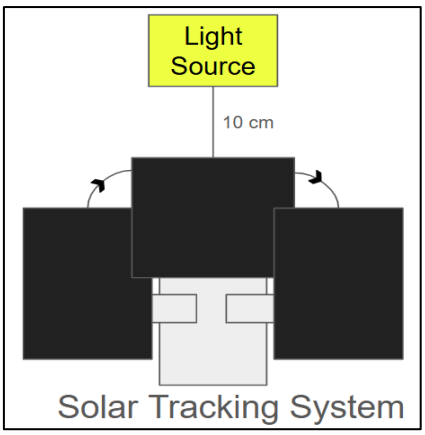

Figure 8 Diagram of Solar Panel movement across Light Source

With the definitions of the tests outlined, the results could be obtained and analysed to conclude the project with respect to the project specifications.

4.2 Results

4.2.I Solar Position Calculation Accuracy

The accuracy of solar position was tested against the solar calculator from the NOAA website. From this testing, data points could be obtained from these calculations and the website to get variation values across the range of test points which the data can be found in a table as Appendix 10. With this table of data, 2 graphs can be formed to show the variation of solar azimuth and elevation across the test dates and time stamps as shown in Figures 9 and 10 respectively.

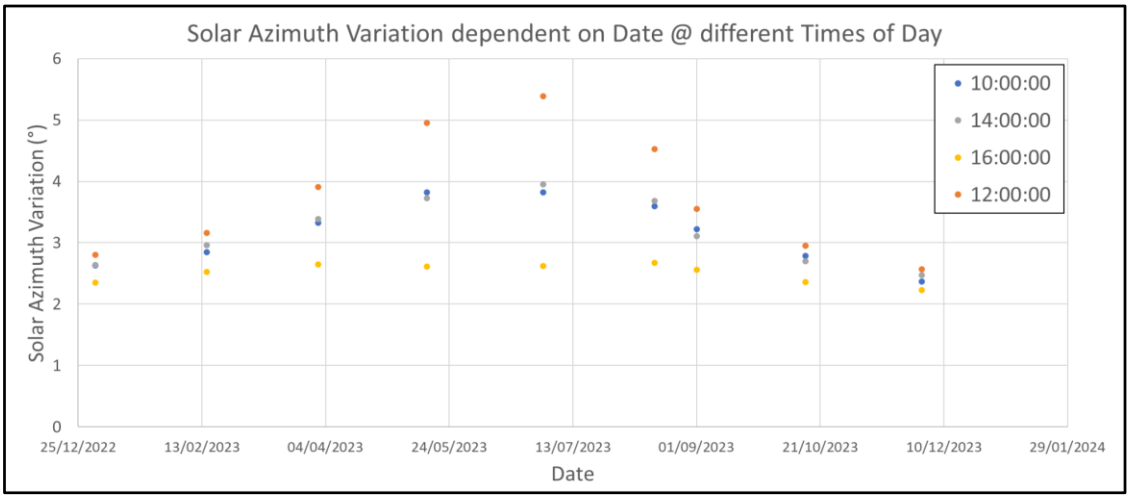

Figure 9 Graph of Solar Azimuth Variation dependent on Date at Different Times of Day

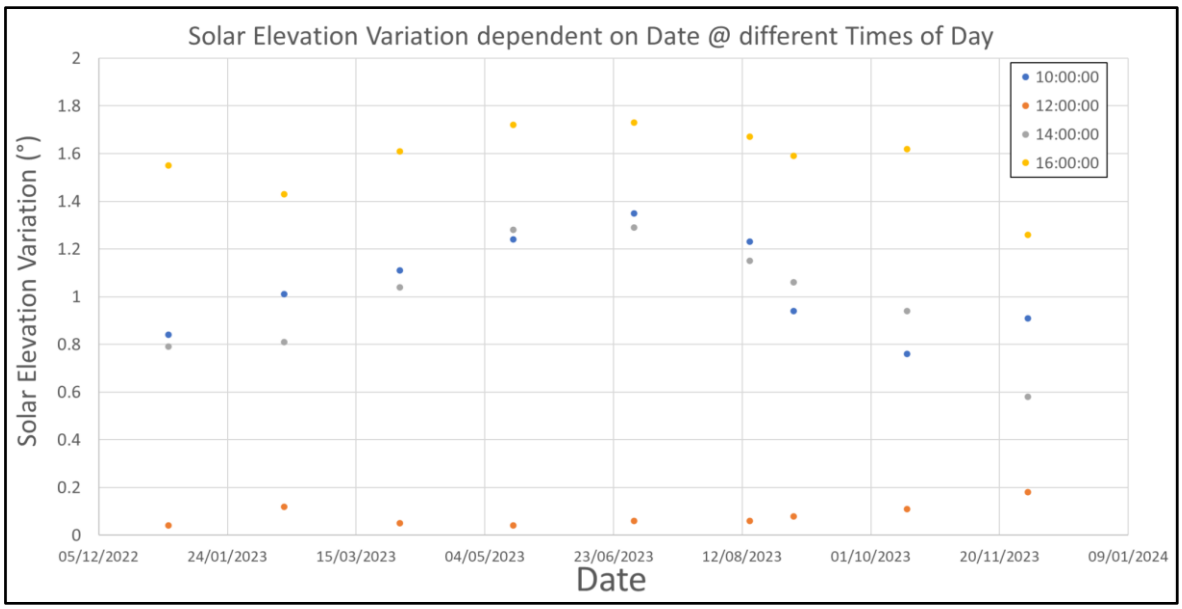

Figure 10 Graph of Solar Elevation Variation dependent on Date at Different Times of Day

Analysing the data, an average variation of the azimuth of 3.37° is found where the azimuth variation only exceeded the 5° specification once on July 1st, 2023, at 12:00 and is found at an average of 0.92° without ever exceeding the 2° specification. From Figure 9, a parabolic trend in variation is observed in the azimuth which increases in late March and decreases in early September which suggests the influence of daylight savings on accuracy which is typically between late March and late October [15]. This trend is most prominent during the timestamps of 10:00, 12:00, and 14:00.

Furthermore, it is observed that the highest average variation of azimuth and elevation is during the hours of 12:00 with 3.87° and 16:00 with 1.57° respectively. This means that the least accurate calculations are obtained during these hours of the day and hold space for improvement when looking to compensate for this inaccuracy such as changing the mathematical algorithm to account for this.

Considering the averages from this data, the low average variation of calculated azimuth and elevation to the standard NOAA calculator suggests a high-accuracy execution of solar position calculations by the system program. The average azimuth and elevation numbers of 3.37° and 0.92° respectively are within their specifications of 5° and 2° and conclude a high

solar position calculation accuracy.

4.2.II Physical Positional Accuracy

Physical positional accuracy was tested, and its data for azimuth and elevation servos are in tables shown in Appendix 11 and 12 respectively. From these tables, 2 graphs can be obtained which depict the variation of physical azimuth and elevation angles against their input angles across their respective ranges as shown in Figures 11 and 12 respectively.

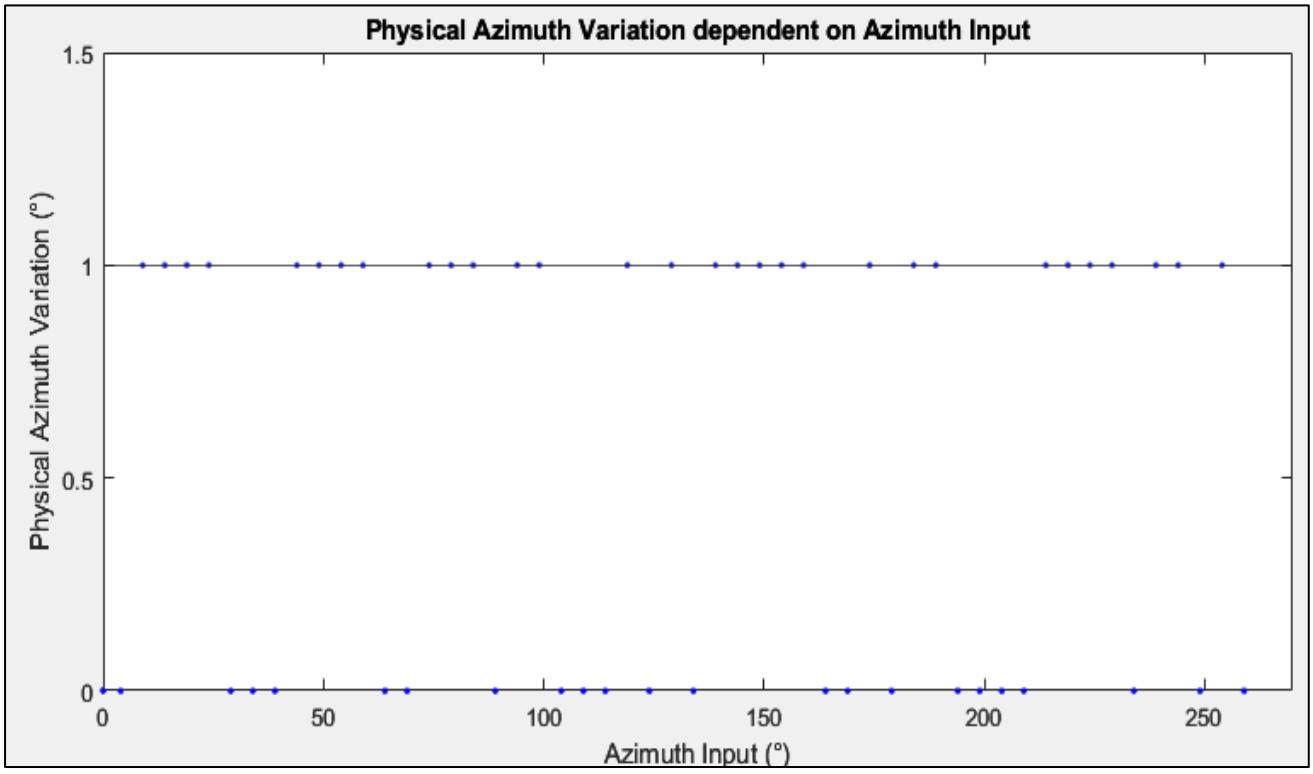

Figure 11 Graph of Physical Azimuth Variation dependent on Azimuth Input

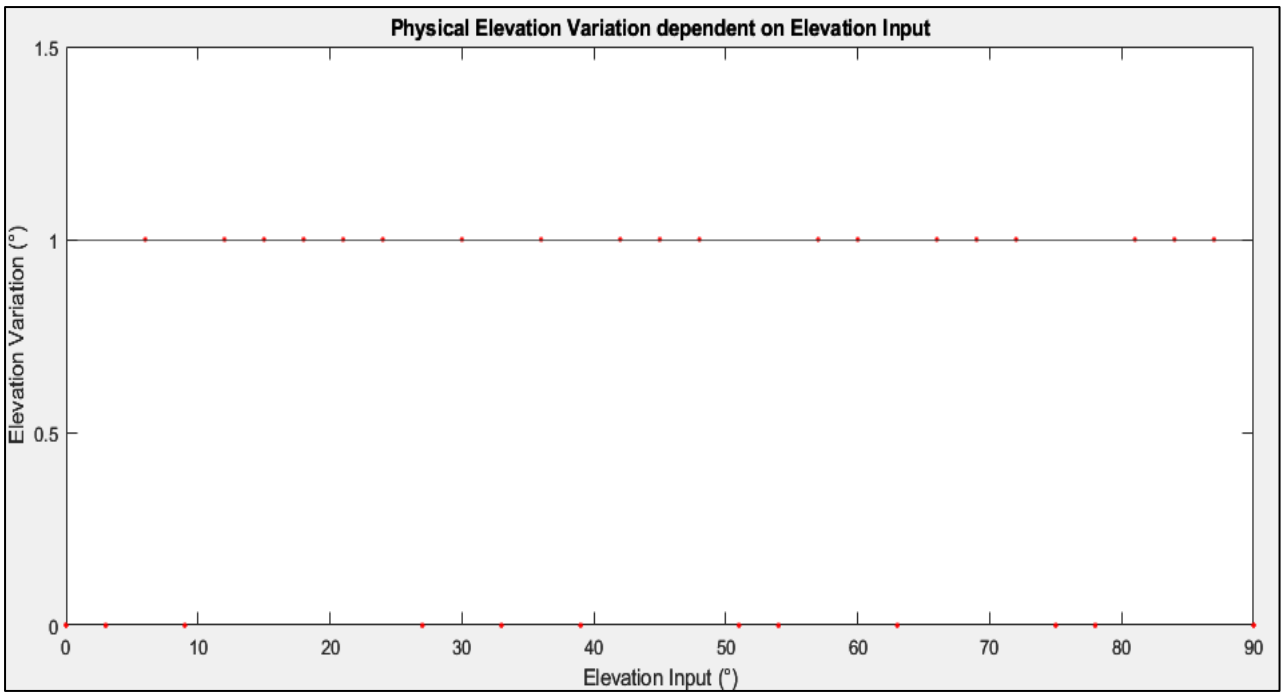

Figure 12 Graph of Physical Elevation Variation dependent on Elevation Input

From Figures 11 and 12, there are no identifiable trends within this testing. However, it is observed that the variations of both the azimuth and elevation are within the physical angle variation specification of 1° across the full range of testing. This is evident as all points are below or on the horizontal line intersecting 1° on the y-axis which represents the specification in both Figures 11 and 12.

Analysing this data, the azimuth sees an average physical variation of 0.57° and elevation shows an average variation of 0.613°. This testing shows an extremely high level of precision in accurately directing the PV panel toward the calculated solar position. Similarly to the solar position calculation testing, it can be concluded from this test that the solar tracking system would have a positive effect on the PV panel's performance.

4.2.III Solar Tracking Performance Effectiveness

Observing the data from the solar tracking performance effectiveness testing, a general understanding of the maximum generation angle range from the direct line of sight from a light source can be obtained. The data from this test is shown in Appendix 13 in a table which is interpreted into a graph as shown in Figure 13.

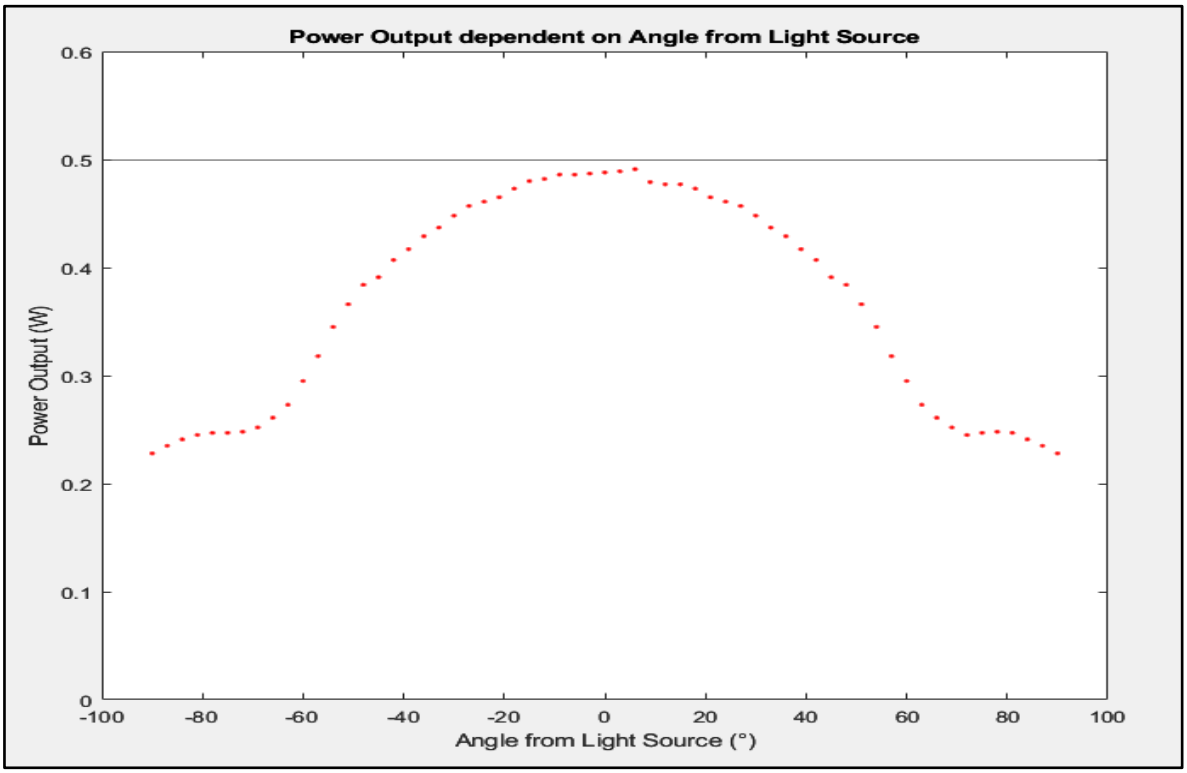

Figure 13 Graph of Power Output dependent on Angle from Line of Sight to Light Source

From a glance, a parabolic trend is shown in Figure 13 which mimics the movement of the PV panel across the static light source as described in Figure 6. The horizontal line intersecting the y-axis at 0.5W of power output helps show the maximum power output of the PV panel and how the power output approaches it as the PV panel approaches the direct line of sight to the light source at 0°.

This test aimed to demonstrate the effect of a moving light source on the power output of a static 0.5W PV panel which would result in an average output power of 0.368W. In comparison, a light tracking PV panel maintaining perfect alignment with the light source would output a continuous 0.5W which would be 26% more power as found by Equation (1).

$$Power\ Increase\ (\%) = 1 - \frac{Static\ Power\ Output}{Maximum/Solar\ Tracking\ Power\ Output} \times 100 \quad (1)$$

Moreover, this test identified the maximum generation angle range as values that were within the 0.475W threshold of maximum power output were found across ±12° from the direct line of sight of the light source. Beyond this ±12° range, the power output of the PV panel exponentially decreases.

To conclude, it is observed that the use of a solar tracking system would help benefit the overall power output of a PV panel by 26% based on this test which exceeds the project specification of 20%. It is also determined that the maximum power output angle range is ±12° from the direct line of sight of the light source.

Chapter 5

5. Conclusions

To conclude the project, 3 phases have held the most significance to the completion of the project that is the design, development, and testing. These 3 phases all had a similar amount of time budgeted for them, with about 9 to 12 weeks allocated for each phase which is outlined by the GANTT chart in Appendix 12.

Beginning with the design phase, 10 weeks were required to complete this phase of the project which allowed for a feasible and complete design as well as suitable tests that would gauge the level of improvement of the PV panel's performance with the solar tracking system. The design of the mathematical model, circuitry, and servo bracket were difficult aspects to design and significant research was required to ensure they were effective in performing their tasks. This phase of the project is hard to conclude without seeing the final product. However, with a fully functional prototype and subsequent iterations, it is proven that the design phase was successful.

Following that, the development phase was the step in which the work from the design phase would be physically constructed and assembled where it could be assessed for their real-life performance. This was a vital role in ensuring that the final product would be able to successfully achieve the targets of the project specification. Throughout this phase, the execution of solar position calculations was the most important aspect to accurately execute as this was a key part of the systems program which ensured the directions towards the Sun were accurate. To ensure all aspects of development were done accurately, they would be checked throughout every addition of commands to minimize large amounts of errors in one setting. This worked well throughout the development of the coded program and the hardware as short circuits were tested throughout the soldering process of the Arduino shield. Furthermore, the 3D-printed servo bracket was tested across its full ranges to ensure that it would maintain its stability. These points illustrate that a well-constructed solar tracking system has been achieved which was ready for the testing phase.

Lastly, the testing phase simulated the final solar tracking system for different aspects of its performance to ensure that the constructed system can perform to the design it is based on.

Throughout this stage, the system aimed to achieve the project specifications laid out in the design phase where the accuracy of solar calculations and physical accuracy and impact of solar tracking were measured. From the solar calculation accuracy testing, it was observed that the solar calculations were accurate as azimuth and elevation had average calculation variations of 3.37° and 0.92° respectively which remained within their respective 5° and 2° specifications. Physical accuracy was also well within its specification of 1° across the full range of testing where averages of azimuth and elevation variations were 0.57° and 0.613° respectively. Solar tracking impact was determined to generate 26% more power in comparison to a static PV panel with a maximum power output angle range of ±12° from the direct line of sight of the light source. The analysis of these 3 tests concludes that the solar tracking system is effective in boosting the performance of the PV panel as the accuracy of the system successfully maintains an angle between the Sun and the PV panel within the angle range which allows maximum power output at all times of the day.

Overall, this dissertation project has concluded that a successful automated dual-axis solar tracking system has been designed, developed, and tested. In terms of achieving the project specifications, all project specifications have been met or exceeded which is a testament to the accuracy of execution of the project.

Considering this success, it is also important to reflect upon the work done how it could have been improved, and what future work can be done. Firstly, the measurement of the physical azimuth and elevation angles were tested with a physical protractor which could not take high decimal measurements. This limitation could have been tackled with an electronic measurement tool such as an electronic angle finder. Secondly, the use of a compass and GPS would have been great additions to the project as it would allow for simple alignment to the North Pole and help create a more robust design. Thirdly, conducting further research on the use and effectiveness of solar tracking systems by considering other designs such as a hybrid sensor and sensorless solar tracking design, and exploring the true efficiency of having a solar tracking system and whether the power that is used to power the system is more or less than what it helps increase in generation. Future work could explore hybrid sensor-algorithm designs for industry applications.

Chapter 6

6. Economic, Legal, Social, Ethical and Environmental Context

An automated dual-axis solar tracking system has a wide range of engineering and external considerations which are important in analysing when evaluating a wider application and development of such systems beyond such an academic study.

Economically, solar tracking systems can potentially increase the power and energy generation of PV systems and improve the overall financial return of such systems. However, the increased control installation may also increase the upfront and maintenance costs may require economies of scale to be successful. Considering legality, the solar tracking system will have many safety and regulatory standards to uphold which is especially true when considering residential integration with unskilled people being at a higher risk of the electrical hazards that may arise from a large-scale solar tracking system. From a societal view, this solar tracking system can help bring accessibility to solar energy to some and be educational to others on how solar energy can be further advanced. Ethical considerations of sourcing, usage, and eventual disposal of electrical and mechanical components will contribute to waste and the overall environmental footprint of the PV system. Environmentally, the system aims to positively affect by increasing reliability in sustainable energy and reducing reliance on fossil fuels. This would help reduce carbon emissions but considering the increased materials of the solar tracking system may offset the environmental benefits it provides. Future assessment could include a life cycle analysis to evaluate environmental impact across product life stages which would both benefit an ethical and environmental perspective.

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Appendix

Appendix 1 Final Code of Arduino Program

```
#include <SPI.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <I2C_RTC.h>
#include <SolarCalculator.h>
#include <TimeLib.h>
#include <Servo.h>

//TUP FOR LCD DISPLAY
#define SCREEN_WIDTH 128 // OLED display width, in pixels
#define SCREEN_HEIGHT 64 // OLED display height, in pixels

#define OLED_RESET     -1 // Reset pin # (or -1 if sharing Arduino reset pin)
#define SCREEN_ADDRESS 0x3C ///< See datasheet for Address; 0x3D for 128x64,
0x3C for 128x32
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, OLED_RESET);

//Controlling servo motors
Servo xAxisServo;  // create Servo object to control a servo
Servo yAxisServo;

int posX = 0;    // variable to store the servo position
int posY = 0;

// Location

  float latitudeDeg = 53;//potentially can change to read fromGPS
  double latitudeMin = 22;//potentially can change to read fromGPS
  double latitudeSec = 54;//potentially can change to read fromGPS
  double longitudeDeg =  1; //potentially can change to read from GPS,
Currently set at arbitrary value
  double longitudeMin = 28; //potentially can change to read from GPS,
Currently set at arbitrary value
  double longitudeSec = 43; //potentially can change to read from GPS,
Currently set at arbitrary value

  //finding the lat and long in degree decimal
  // double latitude  = latitudeDeg + latitudeMin/60 + latitudeSec/3600;
  // double longitude = longitudeDeg + longitudeMin/60 + longitudeSec/3600;
```

```
  double latitude  = 53.38085;
  double longitude = 1.47818;

  int utc_offset = 0;

  //variables of azimuth and elevation
  double az, el;

  //measurement of angle from potentiometer in servo
  float angleTrue;

// Refresh interval, in seconds
int interval = 10;

//SETUP FOR PMOD RTC
#define eepromRead 0xAF   // Binary value is 10101111
#define eepromWrite 0xAE  // Binary value is 10101110
#define sramRead 0xDF     // Binary value is 11011111
#define sramWrite 0xDE    // Binary value is 11011110

uint8_t rgbSec [2] = {0x00, 0x00}; // {Address, Value to be written}, this is
for seconds
uint8_t rgbMin [2] = {0x01, 0x00}; // {Address, Value to be written}, this is
for minutes
uint8_t rgbHour [2] = {0x02, 0x00}; // {Address, Value to be written}, this is
for Hours
uint8_t rgbDay [2] = {0x03, 0x00}; // Days of the week
uint8_t rgbDate [2] = {0x04, 0x00}; // Calander date (1-31)
uint8_t rgbMonth [2] = {0x05, 0x00}; // calander month (1-12)
uint8_t rgbYear [2] = {0x06, 0x00}; // year (0-99)


static MCP7940 RTC; // Setting time Auto

//Defining Variables from RTC
int secondsRTC, minutesRTC, hoursRTC, days;
char writtenDays;
int dateRTC;
int monthRTC;
int yearRTC;

//Values for Voltmeter
double volt;
double power;
int analogVolt;
```

```
//Program Setup
void setup() {
  Wire.begin();
  Serial.begin(9600);

  //Assigning pin numbers for servo control
  xAxisServo.attach(11);  // attaches the servo on pin 9 to the Servo object
  yAxisServo.attach(10);  // attaches the servo on pin 9 to the Servo object


  // SSD1306_SWITCHCAPVCC = generate display voltage from 3.3V internally
  if(!display.begin(SSD1306_SWITCHCAPVCC, SCREEN_ADDRESS)) {
    Serial.println(F("SSD1306 allocation failed"));
    for(;;); // Don't proceed, loop forever
  }

  // Show initial display buffer contents on the screen --
  // the library initializes this with an Adafruit splash screen.
  display.display();

  // Clear the buffer
  display.clearDisplay();

  // Draw a single pixel in white
  display.drawPixel(10, 10, SSD1306_WHITE);

  display.display();
    uint8_t temp;

  rgbSec[0] = 0x00;
  Wire.beginTransmission(0x6F);
  Wire.write(rgbSec[0]);
  Wire.endTransmission();

  // Read from address
  Wire.requestFrom(0x6F, 1);

  // Get second
  while (Wire.available()) {
    temp = Wire.read();
  }

  //set the 7th bit to stop the clock
  temp = (temp | 0x80);

  // //start clock
```

```
  // Wire.beginTransmission(0x6F);
  // Wire.write(0x00);
  // Wire.write(temp);
  // Wire.endTransmission();

  // // Write seconds
  // rgbSec[0] = 0x00;
  // //set address
  // Wire.beginTransmission(0x6F);
  // Wire.write(rgbSec[0]);
  // Wire.endTransmission();

  // // Set Seconds
  // Wire.beginTransmission(sramRead);
  // Wire.write(rgbSec[0]);
  // Wire.write(rgbSec[1]);
  // Wire.endTransmission();

  // Wire.requestFrom(0x6F, 1);
  // temp = Wire.read();

  // if ((temp & 0x80) == 0x80) { //preserve configuration bits
  //   rgbSec[1] = (rgbSec[1] | 0x80);
  // }

  // //set minute
  // rgbMin[1] = 20;  // this is the decimal number (57) that coorsponds to
the hex number (39)
  // Wire.beginTransmission(0x6F); // was 0x6F
  // Wire.write(rgbMin[0]);
  // Wire.write(rgbMin[1]);
  // Wire.endTransmission();

  // //Set Hour
  // rgbHour[0] = 0x02;
  // rgbHour[1] = 18; // 33 is the Decimal number that coorsponds to the Hex
number 21 (9PM);

  // Wire.beginTransmission(0x6F); // was 0x6F
  // Wire.write(rgbHour[0]);
  // Wire.write(rgbHour[1]);
  // Wire.endTransmission();

  // //Set Day of the week (days)
  // rgbDay[0] = 0x03;
  // rgbDay[1] = 0x07;
```

```
// //set address
// Wire.beginTransmission(0x6F);
// Wire.write(rgbDay[0]);
// Wire.endTransmission();

// //read from address
// Wire.requestFrom(0x6F, 1);

// while (Wire.available()) {
//   temp  = Wire.read(); // 0x06 = 6 = Friday
// }

// //reserve the control bits
// rgbDay[1] = ((rgbDay[1] & 0x07) | (temp & 0xF8));

// //set day
// Wire.beginTransmission(0x6F);
// Wire.write(rgbDay[0]);
// Wire.write(rgbDay[1]);
// Wire.endTransmission();

// // set calander date (1-31)
// rgbDate[1] = 0x6; // enter the value for the date here

// //Set address of the date register
// rgbDate[0] = 0x04;

// //set date
// Wire.beginTransmission(0x6F);
// Wire.write(rgbDate[0]);
// Wire.write(rgbDate[1]);
// Wire.endTransmission();

// //set month
// rgbMonth[1] = 0x05; // sets the current month

// //Set address of the month register
// rgbMonth[0] = 0x05;
// //set month
// Wire.beginTransmission(0x6F);
// Wire.write(rgbMonth[0]);
// Wire.write(rgbMonth[1]);
// Wire.endTransmission();

// //Set Year
```

```
  // rgbYear[0] = 0x06;
  // rgbYear[1] = 0x19; // 14 in Hex is 20 in decimal.
  // //set year
  // Wire.beginTransmission(0x6F);
  // Wire.write(rgbYear[0]);
  // Wire.write(rgbYear[1]);
  // Wire.endTransmission();
}


//PRogram being Run Continuously
void loop() {
  //get seconds
  uint8_t bSec;
  uint8_t bSecBuffer = 0;


  bSec = 0x00;

  // Set address
  Wire.beginTransmission(0x6F);
  Wire.write(bSec);
  Wire.endTransmission();

  // Read address
  Wire.requestFrom(0x6F, 1);
  while (Wire.available())    {
    bSecBuffer = Wire.read();
  }

  // Return valid bits
  secondsRTC = bSecBuffer & 0x7F;  // this returns a Decimil value for seconds

  // Get minutes
  uint8_t bMinBuffer = 0;
  uint8_t bMin;

  // RTCC_RTCC
  bMin = 0x01;

  //set address
  Wire.beginTransmission(0x6F);
  Wire.write(bMin);
  Wire.endTransmission();

  //read from address
```

```
Wire.requestFrom(0x6F, 1);

while (Wire.available()) {
  bMinBuffer  = Wire.read();
}

minutesRTC = bMinBuffer & 0x7F;  // returns the decimal value for mintues

// Get hours
uint8_t bHourBuffer = 0;
uint8_t bHour;

//Set address of the hour register
bHour = 0x02;

//set address
Wire.beginTransmission(0x6F);
Wire.write(bHour);
Wire.endTransmission();

//read from address
Wire.requestFrom(0x6F, 1);

while (Wire.available()) {
  bHourBuffer  = Wire.read();
}

hoursRTC = bHourBuffer & 0x3F;

// Get Day
uint8_t bDayBuffer = 0;
uint8_t bDay;

//Set address of the day register
bDay = 0x03;

//set address
Wire.beginTransmission(0x6F);
Wire.write(bDay);
Wire.endTransmission();

//read from address
Wire.requestFrom(0x6F, 1);

while (Wire.available()) {
  bDayBuffer  = Wire.read();
```

```
}

// return valid bits
days = (bDayBuffer & 0x07);



//Get date
uint8_t bDateBuffer = 0;
uint8_t bDate;

bDate = 0x04; //calendar date

//set address
Wire.beginTransmission(0x6F);
Wire.write(bDate);
Wire.endTransmission();

//read from address
Wire.requestFrom(0x6F, 1);

while (Wire.available()) {
  bDateBuffer  = Wire.read();
}

dateRTC = bDateBuffer & 0x3F;

//Get Month
uint8_t bMonthBuffer = 0;
uint8_t bMonth;

bMonth = 0x05;

//set address
Wire.beginTransmission(0x6F);
Wire.write(bMonth);
Wire.endTransmission();

//read from address
Wire.requestFrom(0x6F, 1);

while (Wire.available()) {
  bMonthBuffer  = Wire.read();
}

monthRTC = bMonthBuffer & 0x1F;
```

```
  //Get Year
  uint8_t bYearBuffer = 0;

  //set address
  // only RTCC_RTCC has the year register
  Wire.beginTransmission(0x6F);
  Wire.write(0x06);
  Wire.endTransmission();

  //read from address
  Wire.requestFrom(0x6F, 1);

  while (Wire.available()) {
    bYearBuffer  = Wire.read();
  }

  yearRTC = bYearBuffer & 0xFF;

  setTime(hoursRTC, minutesRTC, secondsRTC, dateRTC, monthRTC, yearRTC);

  static unsigned long next_millis = 0;
  // At every interval
  // if (millis() > next_millis)
  // {
  time_t utc = now();
  //servoReset();
  //delay(5000);
  servoAdjust();
  voltMeasure();
  printValues();
}  ////////////////////////////

void servoReset(){
  xAxisServo.write(0);              // tell servo to go to position in
variable 'pos'
  yAxisServo.write(0);              // tell servo to go to position in
variable 'pos'
}

void servoAdjust(){
  int angleCmd = az * 0.695; //had to change and adjust as the servo was not
recording its angles correctly, had to manually change to fix this issue by
checking the voltage across the potentiometer of servo, measuring angles and
finfind the relative change
  int feedbackValue = analogRead(A2); // 0-1023
```

```
  float voltage = feedbackValue * (5 / 1023.0); // Convert to volts
  angleTrue = (voltage-0.18) / 2.1 * 261;
  if (az <=260){
  xAxisServo.write(180-angleCmd);              // tell servo to go to position
of Azimuth
  yAxisServo.write(el);              // tell servo to go to position of
Elevation
  }
  else if (az >260){
  xAxisServo.write(angleCmd-125.1);              // tell servo to go to
position of Azimuth
  yAxisServo.write(180-el);              // tell servo to go to position of
Elevation
  }
}

//voltmeter function
void voltMeasure(){
  analogVolt = analogRead(A0);
  volt=((double)analogVolt*5.1)/1023;
  power = volt/10;
}

void printValues() {
  // Initialise variables
int cv = SSD1306_WHITE;
int cb = SSD1306_BLACK;

// Display of values
//Settings of size of display print
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(SSD1306_WHITE);
display.setCursor(0,0);

//Printing Date
display.print("Date: ");
// Add Code for Date->
display.print(dateRTC, HEX);
display.print("/");

switch (monthRTC) {
  case 0x01:
    display.print("Jan");
    break;
  case 0x02:
```

```
    display.print("Feb");
    break;
  case 0x03:
    display.print("Mar");
    break;
  case 0x04:
    display.print("Apr");
    break;
  case 0x05:
    display.print("May");
    break;
  case 0x06:
    display.print("Jun");
    break;
  case 0x07:
    display.print("Jul");
    break;
  case 0x08:
    display.print("Aug");
    break;
  case 0x09:
    display.print("Sep");
    break;
  case 0x10:
    display.print("Oct");
  case 0x11:
    display.print("Nov");
  case 0x12:
    display.print("Dec");
    break;
  default:
    display.print("Error");
    break;
  }
display.print("/");
display.print("20");
display.println(yearRTC);

//Printing Time
display.print("Time: ");
//Add code for Time
//->
display.print(" ");
display.print(hoursRTC, HEX);
display.print(":");
if (minutesRTC < 10){
```

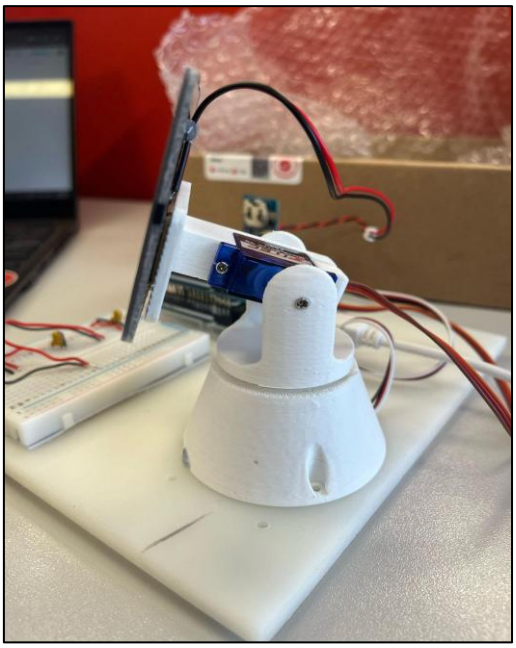
```
  display.print("0");
  display.print(minutesRTC, HEX);
}
else {
  display.print(minutesRTC, HEX);
}
display.print(":");
if (secondsRTC < 10){
  display.print("0");
  display.println(secondsRTC, HEX);
}
else {
  display.println(secondsRTC, HEX);
}

//Values of Coordinates
display.print("Latitude: ");
display.println(latitude);
display.print("Longitude: ");
display.println(longitude);

calcHorizontalCoordinates(yearRTC, monthRTC, decToHex(dateRTC),
decToHex(hoursRTC), decToHex(minutesRTC), decToHex(secondsRTC), latitude,
longitude, az, el);
if (el < 0){
  el = 0 - el;
}
//Servo Reset
//az = 0;
//el = 0;

//Values of Az and El
display.print("Azimuth: ");
display.print(az);
display.println("deg");
display.print("Elevation: ");
display.print(el);
display.println("deg");
display.print("Voltage: ");
display.print(volt);
display.println("V");
display.print("Power: ");
display.print(power);
display.println("W");
display.display();
```

```
Serial.print(F("Az: "));
    Serial.print(az);
    Serial.print(F(" °  El: "));
    Serial.print(el);
    Serial.println(F("°"));
    Serial.print(F(" Hours: "));
    Serial.print(decToHex(hoursRTC));
    Serial.print(F(" Minutes: "));
    Serial.print(decToHex(minutesRTC));
    Serial.print(F(" Seconds: "));
    Serial.print(decToHex(secondsRTC));
    Serial.print(F(" Date: "));
    Serial.print(decToHex(dateRTC));
    Serial.print(F(" month: "));
    Serial.print(monthRTC);
    Serial.print(F(" Year: "));
    Serial.print(yearRTC);
delay(1000);
}

//converting decimal to hexadecimal
int decToHex(int x){
  int rem, count, hex = 0;
  count = x / 16;
  rem = x%16;
  hex += count*10;
  for (int tmp = 1; tmp <= rem; tmp++){
      if (tmp < 10){
        hex += 1;
      }
    }
  return(hex);
}

//Time Library Fuinctions
time_t toUtc(time_t local)
{
  return local - utc_offset * 3600L;
}

// Code from JChristensen/Timezone Clock example
time_t compileTime()
{
  const uint8_t COMPILE_TIME_DELAY = 8;
  const char *compDate = __DATE__, *compTime = __TIME__, *months =
"JanFebMarAprMayJunJulAugSepOctNovDec";
```

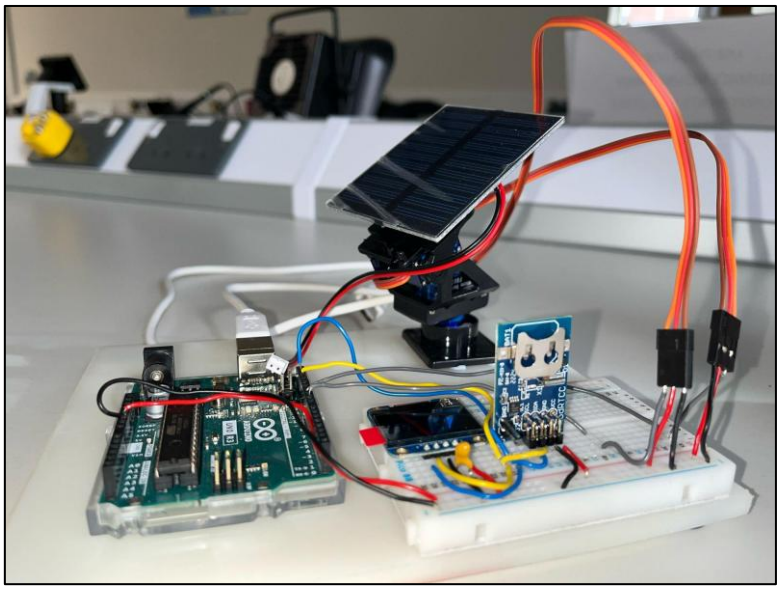
```
  char chMon[4], *m;
  tmElements_t tm;

  strncpy(chMon, compDate, 3);
  chMon[3] = '\0';
  m = strstr(months, chMon);
  tm.Month = ((m - months) / 3 + 1);

  tm.Day = atoi(compDate + 4);
  tm.Year = atoi(compDate + 7) - 1970;
  tm.Hour = atoi(compTime);
  tm.Minute = atoi(compTime + 3);
  tm.Second = atoi(compTime + 6);
  time_t t = makeTime(tm);
  return t + COMPILE_TIME_DELAY;
}
```

Appendix 2: Picture of Initial Servo Bracket

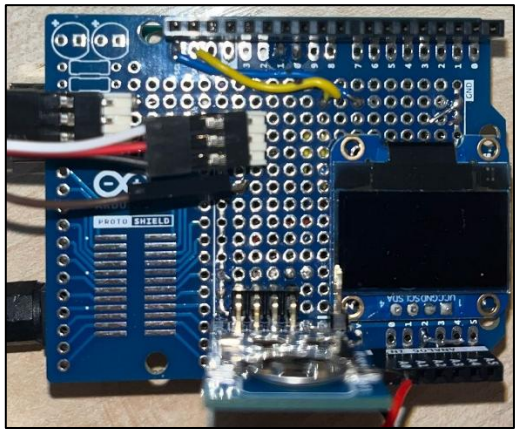
Appendix 3: Picture of Final 3D Printed Servo Bracket

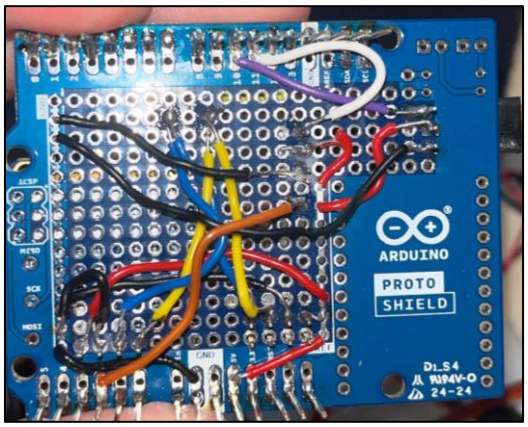


Appendix 4 Breadboard Circuitry

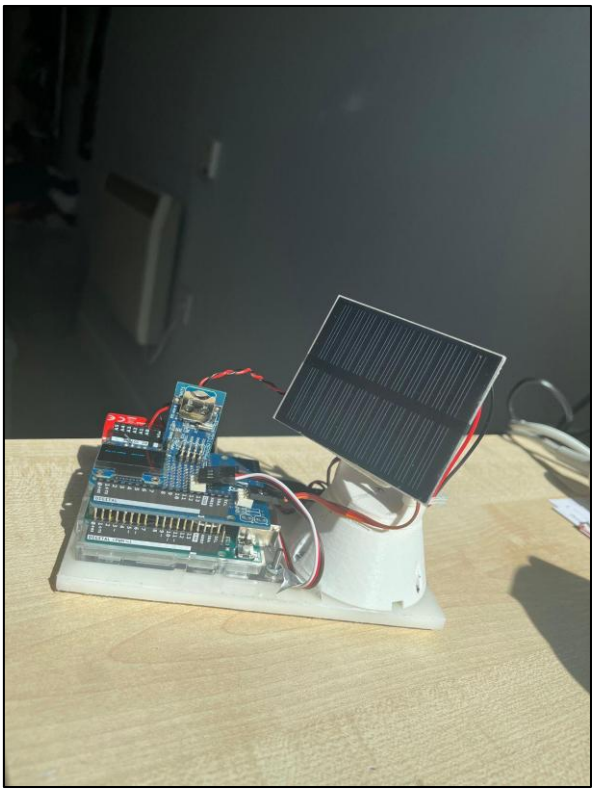


Appendix 5 Proto Shield with Full Circuitry

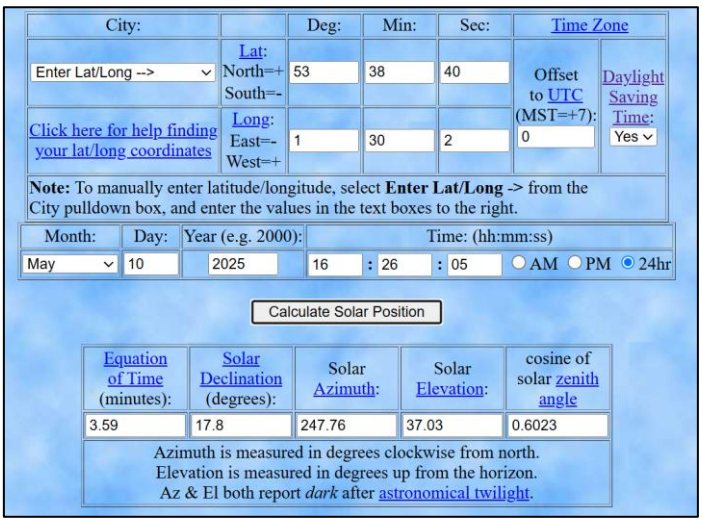




Appendix 6 Assembled Solar Tracking System



Appendix 7 NOAA Solar Calculator: https://gml.noaa.gov/grad/solcalc/azel.html



City:		Deg:	Min:	Sec:	Time Zone	
Enter Lat/Long -->	Lat: North=+ South=-	53	38	40	Offset to UTC (MST=+7): 0	Daylight Saving Time: Yes
Click here for help finding your lat/long coordinates	Long: East=- West=+	1	30	2		

Note: To manually enter latitude/longitude, select **Enter Lat/Long** -> from the City pulldown box, and enter the values in the text boxes to the right.

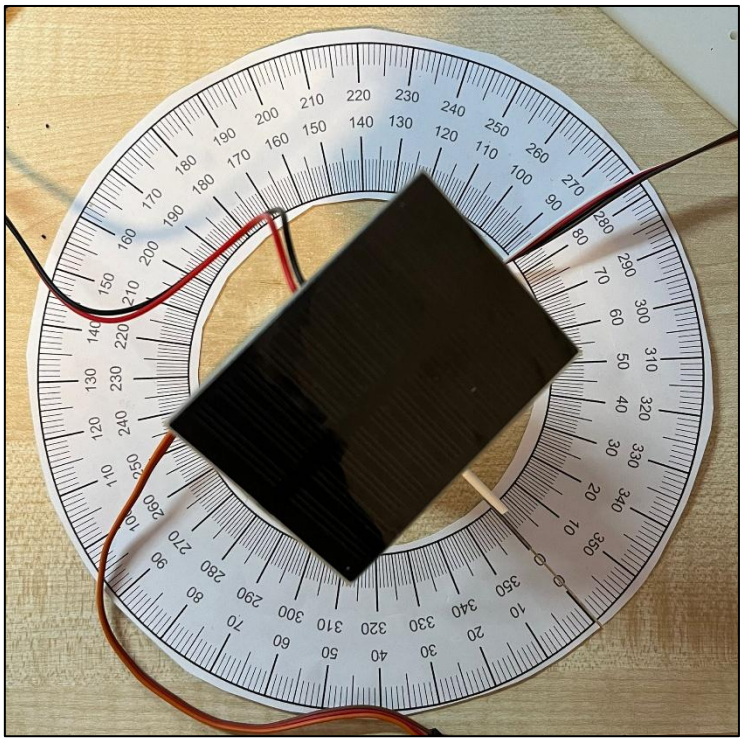
Month:	Day:	Year (e.g. 2000):	Time: (hh:mm:ss)			
May	10	2025	16 :	26 :	05	AM PM 24hr

Calculate Solar Position

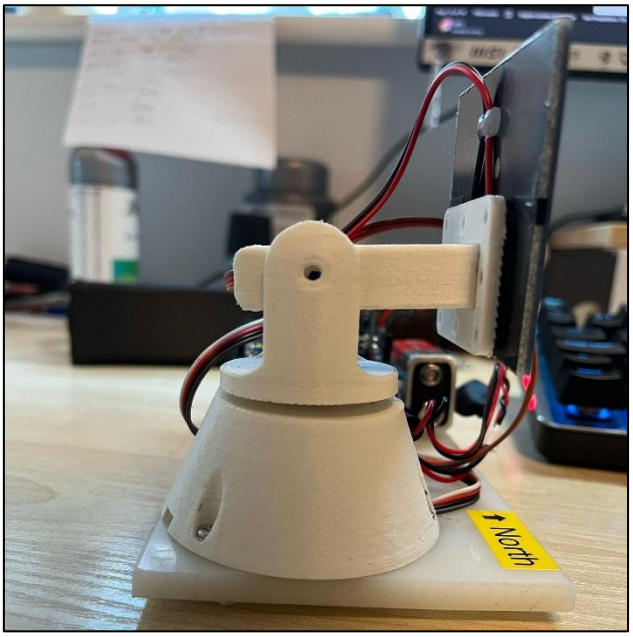
Equation of Time (minutes):	Solar Declination (degrees):	Solar Azimuth:	Solar Elevation:	cosine of solar zenith angle
3.59	17.8	247.76	37.03	0.6023

Azimuth is measured in degrees clockwise from north.
Elevation is measured in degrees up from the horizon.
Az & El both report *dark* after astronomical twilight.

Appendix 8 Protractor set up for Azimuth



Appendix 9 Protractor set up for Elevation



Appendix 10 Table of Data for Solar Position Calculation Accuracy

Time (HH:MM:SS)	Date (DD/MM/YYYY)	Solar Azimuth (NOAA) (°)	Solar Elevation (NOAA) (°)	Solar Azimuth (°)	Solar Elevation (°)	Solar Azimuth Variation (°)	Solar Elevation Variation (°)	Average Solar Azimuth Variation (per timestamp)	Average Solar Elevation Variation (per timestamp)
10:00:00	01/01/2023	150.13	8.74	152.76	9.58	2.63	0.84		
10:00:00	15/02/2023	144.07	17.51	146.92	18.52	2.85	1.01		
10:00:00	01/04/2023	139.57	34.41	142.9	35.52	3.33	1.11		
10:00:00	15/05/2023	133.78	48.2	137.6	49.44	3.82	1.24		
10:00:00	01/07/2023	128.25	51.09	132.07	52.44	3.82	1.35	3.158888889	1.043333333
10:00:00	15/08/2023	134.32	43.04	137.92	44.27	3.6	1.23		
10:00:00	01/09/2023	146.66	28.51	149.88	29.45	3.22	0.94		
10:00:00	15/10/2023	153.02	14.37	155.81	15.13	2.79	0.76		
10:00:00	01/12/2023	150.42	8.66	152.79	9.57	2.37	0.91		
12:00:00	01/01/2023	177.78	13.65	180.58	13.69	2.8	0.04		
12:00:00	15/02/2023	174.65	23.84	177.81	23.96	3.16	0.12		
12:00:00	01/04/2023	176.74	41.14	180.65	41.19	3.91	0.05		
12:00:00	15/05/2023	179.04	55.49	183.99	55.45	4.95	0.04		
12:00:00	01/07/2023	175.55	59.67	180.94	59.73	5.39	0.06	3.756666667	0.082222222
12:00:00	15/08/2023	175.99	50.61	180.52	50.67	4.53	0.06		
12:00:00	01/09/2023	181.29	33.43	184.84	33.35	3.55	0.08		
12:00:00	15/10/2023	182.38	18.15	185.33	18.04	2.95	0.11		
12:00:00	01/12/2023	178.04	13.49	180.61	13.67	2.57	0.18		
14:00:00	01/01/2023	205.7	10.04	208.34	9.25	2.64	0.79		
14:00:00	15/02/2023	206.12	20.64	209.08	19.83	2.96	0.81		
14:00:00	01/04/2023	214.86	36.26	218.25	35.22	3.39	1.04		
14:00:00	15/05/2023	224.79	48.72	228.52	47.44	3.73	1.28		
14:00:00	01/07/2023	225.33	53.26	229.28	51.97	3.95	1.29	3.181111111	0.993333333
14:00:00	15/08/2023	219.21	45.1	222.89	43.95	3.68	1.15		
14:00:00	01/09/2023	215.6	27.72	218.71	26.66	3.11	1.06		
14:00:00	15/10/2023	211.4	12.94	214.1	12	2.7	0.94		
14:00:00	01/12/2023	205.89	9.81	208.36	9.23	2.47	0.58		
16:00:00	01/01/2023	231.05	-0.83	233.4	-2.38	2.35	1.55		
16:00:00	15/02/2023	234.05	9.22	236.58	7.79	2.53	1.43		
16:00:00	01/04/2023	245.61	22.59	248.26	20.98	2.65	1.61		
16:00:00	15/05/2023	256.68	33.19	259.29	31.47	2.61	1.72		
16:00:00	01/07/2023	258.19	37.55	260.81	35.82	2.62	1.73	2.508888889	1.575555556
16:00:00	15/08/2023	251.37	30.41	254.04	28.74	2.67	1.67		
16:00:00	01/09/2023	244.26	14.07	246.82	12.48	2.56	1.59		
16:00:00	15/10/2023	237.16	0.83	239.52	-0.79	2.36	1.62		
16:00:00	01/12/2023	231.18	-1.15	233.41	-2.41	2.23	1.26		
Average Variation (°)						3.15	0.92		

Appendix 11 Table of Data for Azimuth Physical Positional Accuracy

Azimuth Input (°)	Azimuth Physical Measurement (°)	Physical Azimuth Variation (°)
0	0	0
4	4	0
9	10	1
14	15	1
19	20	1
24	25	1
29	29	0
34	34	0
39	39	0
44	45	1
49	50	1
54	55	1
59	60	1
64	64	0
69	69	0
74	75	1
79	80	1
84	85	1
89	89	0
94	95	1
99	100	1
104	104	0
109	109	0
114	114	0
119	120	1
124	124	0
129	130	1
134	134	0
139	140	1
144	145	1
149	150	1
154	155	1
159	160	1
164	164	0
169	169	0
174	175	1
179	179	0

184	185	1
189	190	1
194	194	0
199	199	0
204	204	0
209	209	0
214	215	1
219	220	1
224	225	1
229	230	1
234	234	0
239	240	1
244	245	1
249	249	0
254	253	1
259	259	0
Average Physical Azimuth Variation (°)		0.57

Appendix 12 Table of Data for Elevation Physical Positional Accuracy

Elevation Input (°)	Elevation Measurement (°)	Elevation Variation (°)
0	0	0
3	3	0
6	7	1
9	9	0
12	13	1
15	16	1
18	17	1
21	22	1
24	25	1
27	27	0
30	31	1
33	33	0
36	35	1
39	39	0
42	41	1
45	44	1
48	47	1
51	51	0
54	54	0
57	56	1
60	61	1
63	63	0
66	67	1
69	68	1
72	73	1
75	75	0
78	78	0
81	80	1
84	85	1
87	88	1
90	90	0
Average Physical Elevation Variation (°)		0.613

Appendix 13 Table of Data for Solar Tracking Performance Effectiveness

Angle from Light Source ()	Voltage (V)	Power (W)
90	2.28	0.228
87	2.35	0.235
84	2.41	0.241
81	2.47	0.247
78	2.48	0.248
75	2.47	0.247
72	2.45	0.245
69	2.52	0.252
66	2.61	0.261
63	2.73	0.273
60	2.95	0.295
57	3.18	0.318
54	3.45	0.345
51	3.66	0.366
48	3.84	0.384
45	3.91	0.391
42	4.07	0.407
39	4.17	0.417
36	4.29	0.429
33	4.37	0.437
30	4.48	0.448
27	4.57	0.457
24	4.61	0.461
21	4.65	0.465
18	4.73	0.473
15	4.77	0.477
12	4.77	0.477
9	4.79	0.479
6	4.91	0.491
3	4.89	0.489
0	4.88	0.488
-3	4.87	0.487
-6	4.86	0.486
-9	4.86	0.486
-12	4.82	0.482
-15	4.8	0.48
-18	4.73	0.473
-21	4.65	0.465
-24	4.61	0.461
-27	4.57	0.457

-30	4.48	0.448
-33	4.37	0.437
-36	4.29	0.429
-39	4.17	0.417
-42	4.07	0.407
-45	3.91	0.391
-48	3.84	0.384
-51	3.66	0.366
-54	3.45	0.345
-57	3.18	0.318
-60	2.95	0.295
-63	2.73	0.273
-66	2.61	0.261
-69	2.52	0.252
-72	2.48	0.248
-75	2.47	0.247
-78	2.47	0.247
-81	2.45	0.245
-84	2.41	0.241
-87	2.35	0.235
-90	2.28	0.228

Appendix 14 Table of Datasheets

Component Type	Datasheet Link
Servo Drives	https://uk.rs-online.com/web/p/stem-motion-components/2153180?gb=s
LCD Display	https://uk.rs-online.com/web/p/oled-displays/2543586?gb=s
Real Time Clock	https://uk.rs-online.com/web/p/clock-timer-development-tools/1840476?gb=s
Microcontroller	https://uk.rs-online.com/web/p/arduino/2662937?gb=s
Proto Shield - Arduino Uno	https://uk.rs-online.com/web/p/shields-for-arduino/1763652?gb=s
Solar Panel	313070004 | Seeed Studio 0.5W Kit solar panel | RS